Krishna Kant Sharma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.41
Baluni Public School, Agra	Class XII (CBSE)	2020	86.20%
Baluni Public School, Agra	Class X (CBSE)	2018	86.33%

TECHNICAL SKILLS

- **Simulation & Design Tools:** MATLAB/Simulink, IonMonger (drift-diffusion PV solver)
- **Machine Learning & Data Analysis:** scikit-learn (Random Forest), NumPy, SciPy, pyDRTtools (Distribution of Relaxation Times analysis), cross-validation & evaluation design, Python
- **Embedded Systems & IoT:** Arduino UNO, NodeMCU (ESP8266), Blynk IoT Platform, sensor interfacing (soil moisture, ACS712 current sensor, optocouplers), relay/circuit-driver design, I2C-LCD interfacing
- **Power & Energy Domain:** Electrochemical Impedance Spectroscopy (EIS), Power System Analysis & Protection, Power Electronics & Drives, Photovoltaic (PV) System Design, Battery/Energy Storage Characterization

TRAINING AND EXPERIENCE

Industry Research Collaborator - *Pavakah Energy*

(Jul 2026 - Present)

- Collaborating with **Pavakah Energy**, a thin-film photovoltaic startup incubated at TIP, IIT Hyderabad, to validate simulation-based thesis findings through planned hands-on fabrication of perovskite solar cells as the next phase of the research.

Practical Trainee - *RPS Sagar Hydro Power Station, RVUNL, Rawatbhata*

(Jun - Jul 2023)

- Gained hands-on exposure to hydroelectric power generation, studying turbine-generator units, excitation systems, and grid synchronization procedures under the O&M division.
- Observed control room operations, load monitoring, and generator protection schemes, developing a working understanding of hydro plant operation and maintenance practices.
- Performance rated "Excellent" for sincerity and interest in learning new systems.

Practical Trainee - *132kV GSS Mahaveer Nagar, RVPN, Kota*

(May - Jul 2024)

- Studied 132kV grid substation equipment and operations, including power transformers, circuit breakers, isolators, and busbar protection schemes under RVPN's Assistant Engineer.
- Developed practical understanding of substation switching operations, protection relay coordination, and power transmission network management.

PROJECTS

Physics-Guided Machine Learning for Impedance-Based State-of-Health Estimation and Degradation-Mechanism Classification in Perovskite Solar Cells *(M.Tech Thesis)*

Energy Science and Technology, IIT Hyderabad (Jul 2026 - Present)

- Simulated perovskite solar cell degradation (SRH recombination, ion vacancy density) using **IonMonger 2.0**, and extracted an **11-dimensional DRT feature set** from ~400 simulated impedance spectra.
- Trained **Random Forest** models achieving R^2 = 0.996 for state-of-health regression and **99% ± 2% accuracy** for degradation-mechanism classification, using a leakage-free honest-split evaluation methodology.
- Validated results against **58 published studies**, with independent data points matching literature within 0.05-0.13 percentage points.

IoT-Based Prepaid Energy Meter & Home Automation *(B.Tech Major Project)*

- Developed a **NodeMCU (ESP8266)**-based prepaid energy metering system integrated with the **Blynk IoT** app for remote monitoring and appliance control.
- Used an **ACS712 current sensor** for real-time power consumption measurement and automatic unit-based billing.
- Implemented automatic supply cutoff on zero balance, low-balance alerts, and remote recharge functionality via mobile app.

Smart Irrigation System *(B.Tech Minor Project)*

- Designed an **IoT-based automatic irrigation system** using **Arduino UNO** and a soil moisture sensor, triggering a water pump on threshold-based dry-soil detection.
- Built a custom **DC motor driver circuit** (diode-transistor-resistor combination) as a low-cost alternative to a relay module; integrated an **I2C LCD** for real-time moisture and pump status display.

POSITION OF RESPONSIBILITY

Coordinator, Education Program Management - *Prayas (RDC), IIT Hyderabad*

(2025 - 2026)

- Lead weekly volunteer mentoring visits to nearby government schools, supporting students preparing for scholarship exams such as NMMS, and coordinate health & hygiene awareness sessions at a local orphanage.
- Contributed to organizing Prerna, Prayas's flagship outreach event connecting government school students and parents with IAS/IPS/IRS officers and professors for mentorship and campus exposure.

Manager, Startup Partnerships & Outreach (SPO) - *E-Cell, IIT Hyderabad*

(2026-Present)

- Part of the Startup Partnerships & Outreach team, which connects E-Cell with the startup ecosystem by forging partnerships with founders, investors, and organizations for the IIT Hyderabad entrepreneurial community.

EXTRACURRICULAR ACTIVITIES AND PROFESSINAL SKILLS

Event Host / Anchor - *IIT Hyderabad*

- Hosted and anchored multiple college-level events, managing stage presence, audience engagement, and live event flow.

Interests & Hobbies: Cooking, Badminton & Writing

Professional Skills: Communication & Leadership, Analytical & Problem Solving, Time Management & Adaptability, Cross-Functional Collaboration